BSc Thesis Task Description

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Simulation framework for the comparison of quantum surface codes

Quantum computing holds the promise of fundamentally revolutionizing computation by leveraging quantum mechanical phenomena such as superposition, entanglement, and quantum interference. Quantum bits are highly sensitive to environmental interactions as well as noise inherent to hardware control operations and gate implementations. Additionally, unlike classical bits, quantum bits reside in a continuous space, they can exhibit not only bit flip errors, but also phase errors and rotations, making noise mitigation a far more complex challenge.
The field of Quantum Error Correction (QEC) aims to address this challenge, by mapping the information of a single logical qubits into multiple physical qubits. This way, it is possible to handle the unwanted effects of noise, by utilizing redundancy and spatial entanglement, thus allowing the future construction of fault-tolerant quantum computers. Multiple different quantum error correcting code schemes exists, with differing advantages and disadvantages. From these, the family of surface codes is often regarded as the most practical path toward building fault-tolerant, large-scale quantum computers.
The task of the student is to create a simulation framework for these codes, and utilize it to investigate them.
Tasks to be performed by the student will include:

- Review the relevant literature and present the topic of quantum computation and computers, highlighting the main differences, advantages and disadvantages compared to classical counterparts!
- Introduce the theoretical background of quantum error correcting codes, surface codes, describing the operation of the XZZX and CSS codes in particular!
- Based on relevant literature, define the main evaluation criteria, existing options and capabilities for investigating and simulating XZZX and CSS codes!
- Design and implement a simulation framework that can be used to simulate, investigate, and evaluate XZZX and CSS quantum error correcting codes.
- Verify the correct operation of the created simulation framework and evaluate the results!

Supervisor at the department: Dr. Balazs Solymos, associate professor
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